

and the time-reversal symmetry

$$\mathcal{F}_t \circ \mathbf{R} \circ \mathcal{F}_t = \mathbf{R}$$

have a formal similarity, but they are essentially unrelated. A method can be symplectic but not time-reversible (e.g. Symplectic Euler) or it can be time-reversible and not symplectic (e.g. Trapezoidal Rule). However there are intriguing relationships between these properties which we discuss here.

Interestingly the time-reversal symmetry provides analogous properties in some respects to the properties of Hamiltonian systems. For example, consider a linear Hamiltonian system $dz/dt = \mathbf{J}\mathbf{A}z$, with $\mathbf{A}$ a symmetric matrix. If λ is an eigenvalue of $\mathbf{J}\mathbf{A}$ then $\mathbf{J}\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$ for some eigenvector $\mathbf{u} \neq 0$. Because the matrix $\mathbf{J}\mathbf{A}$ is real, we know that $\bar{\lambda}$ will also be an eigenvalue. At the same time, we know that since λ is an eigenvalue of $\mathbf{J}\mathbf{A}$ it is also an eigenvalue of its transpose $(\mathbf{J}\mathbf{A})^T = \mathbf{A}^T\mathbf{J}^T = -\mathbf{A}\mathbf{J}$, thus

$$-\mathbf{A}\mathbf{J}\mathbf{u} = \lambda\mathbf{u}$$

multiplying by $\mathbf{J}$ and setting $\mathbf{v} = \mathbf{J}\mathbf{u}$ we have

$$-\mathbf{J}\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

implying that $-\lambda$ (and hence also $-\bar{\lambda}$) is an eigenvalue of $\mathbf{J}\mathbf{A}$. Real eigenvalues of $\mathbf{J}\mathbf{A}$ are paired with their negatives. If the imaginary part is nonzero, the eigenvalues occur in quadruplets $\{\pm\lambda, \pm\bar{\lambda}\}$. The flow map is

$$\mathcal{F}_t(\mathbf{z}) = e^{t\mathbf{J}\mathbf{A}}\mathbf{z}$$

and the exponential matrix will inherit a related structure within the spectrum: λ an eigenvalue of $\exp(t\mathbf{J}\mathbf{A})$ implies that $\bar{\lambda}$, $1/\lambda$ and $1/\bar{\lambda}$ are all eigenvalues of $\exp(t\mathbf{J}\mathbf{A})$. This eigenvalue structure is generic for linear symplectic maps in general.

Suppose now we have a linear time-reversible map $\Phi(\mathbf{z}) = \mathbf{T}\mathbf{z}$, then

$$\mathbf{T}^{-1} = \mathbf{R}\mathbf{T}\mathbf{R}.$$

Given an eigenvalue, eigenvector pair $(\lambda, \mathbf{u})$ of $\mathbf{T}$, let $\mathbf{u} = \mathbf{R}\mathbf{v}$, so that $\mathbf{R}\mathbf{u} = \mathbf{R}^2\mathbf{v} = \mathbf{v}$, then

$$\mathbf{T}^{-1}\mathbf{v} = \mathbf{R}\mathbf{T}\mathbf{R}\mathbf{v} = \mathbf{R}\mathbf{T}\mathbf{u} = \lambda\mathbf{R}\mathbf{u} = \lambda\mathbf{v}.$$

Thus λ is an eigenvalue of $\mathbf{T}^{-1}$ which, in turn, implies that $1/\lambda$ is an eigenvalue of $\mathbf{T}$. The matrix being real implies that the conjugates of λ and $1/\lambda$ are also eigenvalues, thus we have the same eigenvalue quadruplets as for a linear symplectic map.

Recall that a pair of maps Φ and Ψ are said to be conjugate if there is a homomorphism χ such that